HR EMPLOYEE ATTRITION ANALYSIS PROJECT

- **Project Overview**

This project focuses on analyzing HR Attrition data to understand employee turnover patterns and key influencing factors. The objective is to use data visualization and analysis techniques to derive meaningful insights that help organizations make informed decisions. The Power BI dashboard provides an interactive view of attrition trends across different employee segments.

- **Data Source**

The dataset used in this project is an HR Attrition dataset stored in Excel format. It includes employee details such as age, gender, department, job role, salary, experience, and attrition status. This dataset helps in analyzing workforce behavior and identifying patterns behind employee exits.

- **Tools & Technologies**

- Microsoft Excel – Data source
- Power BI – Data visualization and dashboard creation
- Power Query – Data cleaning and transformation
- DAX – Calculated columns and measures

- **Data Cleaning & Preparation**

The dataset was cleaned and prepared to ensure accuracy and consistency before analysis:

- Removed duplicate records to maintain data integrity
- Handled missing/null values to avoid incorrect insights
- Standardized categorical values (Gender, Department, Attrition)

- Converted data types (numeric, text, date) appropriately
- Renamed columns for better readability
- Removed unnecessary columns to improve performance
- Created calculated columns like Attrition Count and Attrition Rate

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- **Data Modeling**

- Data modeling was performed in Power BI to establish relationships and improve analysis efficiency:

- Defined relationships between tables (if multiple tables used)
- Maintained proper cardinality (one-to-many relationships)
- Created a structured data model for better performance
- Used fact and dimension concepts for organizing data
- Enabled efficient filtering and slicing across visuals

- **Exploratory Data Analysis (EDA)**

EDA was conducted to explore patterns and trends in the dataset:

- Analyzed overall attrition rate across the organization
- Compared attrition across departments and job roles
- Studied the impact of age and experience on attrition
- Evaluated gender-wise attrition distribution
- Used slicers for dynamic filtering and deeper insights
- Visualized data using bar charts, pie charts, and line charts

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- **Key Insights**

- High attrition is observed in certain departments.
  This indicates possible workload issues or lack of growth opportunities.

- Employees with lower experience show higher attrition rates.
  This suggests the need for better onboarding and support.
- Younger employees tend to leave more frequently.
  They are more likely to switch jobs for better opportunities.
- Salary has a significant impact on attrition.
  Lower salary groups experience higher employee turnover.
- Certain job roles have consistently high attrition.
  This may be due to repetitive work or low satisfaction levels.
- Attrition trends vary across different time periods.
  This helps identify peak periods of employee exits.
- Gender has minimal impact on attrition trends.
  Job-related factors play a more important role.

- **Problem Statement**

- Employee attrition is a major challenge for organizations as it affects productivity, increases hiring costs, and impacts overall performance. This project aims to analyze HR data to identify key factors influencing employee attrition and provide actionable insights to improve employee retention and workforce stability.

- **Types of Analysis**

1. Descriptive Analysis

Provides a summary of employee data and overall attrition trends.
Helps identify which departments and groups have higher attrition.

2. Diagnostic Analysis

Analyzes factors like salary, job role, and experience to find reasons for attrition.
Helps understand why employees are leaving the organization.

3. Predictive Analysis

Uses historical patterns to estimate future attrition trends.
Helps identify employees who are at risk of leaving.

4. Prescriptive Analysis

Provides actionable recommendations to reduce attrition.
Supports decision-making to improve employee retention.

- **Recommendations**

- Focus on high-attrition departments with targeted strategies
- Improve salary structure and employee benefits
- Provide clear career growth opportunities
- Enhance employee engagement programs
- Offer training and skill development initiatives
- Monitor attrition trends regularly using dashboards

- **Conclusion**

- This project provides valuable insights into employee attrition and its key influencing factors. The analysis highlights that salary, experience, and job role play a major role in employee turnover.
- By using data visualization and analytical techniques, organizations can identify high-risk areas and take proactive measures to improve retention.
- Overall, this project demonstrates how data-driven decision-making can help organizations reduce attrition and build a more stable workforce